

CCI xG Testbed: An O-RAN Based Platform for Future Wireless Network Experimentation

Aloizio da Silva, Mayukh Roy Chowdhury, Aditya Sathish, Asheesh Tripathi, Scott F. Midkiff, and Luiz A. da Silva

ABSTRACT

The Commonwealth Cyber Initiative (CCI) xG Testbed is a research and experimentation platform that embraces the open-source ideology to enable real-world testing across indoor and outdoor scenarios for next generation (NextG) wireless communications systems. The testbed is an open-access virtualized cloud infrastructure that enables scalable access to Software Defined Radios (SDRs) through its state-of-the-art cloud computing capabilities that leverage software-defined networking and network function virtualization to present an easy-to-use interface to manage radio, compute, and vector processing resources on-demand. In this article, we describe the overall architecture of the CCI xG Testbed, the different sites connected to form a common open, remote-access platform. We also elaborate on how this testbed fits into the overall goals of building open and intelligent RAN as per the ideals of the Open Radio Access Network (O-RAN) Alliance. We present two sample experiments to showcase how the testbed can connect fundamental research with experimentation and real-world testing. We also highlight the gaps the testbed can potentially fill as an enabler for wide-scale system-level research across the NextG Radio Access Network (RAN) stack.

INTRODUCTION

As we progress towards the visioning and standardization of 6G mobile systems, wireless networks are going through significant evolution and technical advancements. Given the diverse use cases, from metaverse to extended reality and automated cars, envisioned to be supported by 6G, the NextG of wireless networks will be AI-native and continue to embrace the trends of disaggregation and softwareization. In this context, Open Radio Access Network (O-RAN) as a technology has initiated a significant paradigm shift in how traditional cellular networks work, revolving around the four pillars of intelligence, openness, virtualization, and disaggregation. The wireless community is in high need of large-scale testbeds that can be used to run experiments for performance evaluation and real-world validation of the technologies being developed in this domain, which are otherwise validated only through simulation. Initiatives such as O-RAN Alliance and 3GPP focus on creating specifications and standards for new and emerging technolo-

gies. Experimentation and testing platforms have a fundamental role to play by implementing, validating, providing guidelines to those initiatives for specifications enhancement and thereby influencing the standards.

The NEXTG O-RAN provides an open, programmable, interoperable, and virtualized RAN architecture which empowers network operators to seamlessly integrate hardware and software components from different vendors, resulting in heightened flexibility, scalability, and cost-efficiency. At the center of O-RAN are open, programmable, and virtualized network components, connected to each other through open interfaces that enable closed-loop, data-driven intelligent control. While the open architecture, softwareization, virtualization and integration of AI/ML hold the promise of transformation, these very features of O-RAN require continuous Testing and Evaluation (T&E) to support a diversified supply chain for wireless networks, spur innovation, and integrate AI into the network in effective ways. To address these issues comprehensively, rigorous analysis, testing, and experimentation on a real-world, NEXTG research infrastructure platform are imperative. Such a platform may consider the following features and capabilities. These desirable features were distilled from discussions held in the IEEE International Network Generations Roadmap (INGR) Roadmap (Testbed Working Group (WG)), DoD National Spectrum Consortium (NSC) (O-RAN WG) and O-RAN Alliance next Generation Research Group (nGRG).

- **Medium-to-Large Scale Geographic Distribution.** Testbeds must be capable of operating at a medium-to-large scale across geographically distributed locations. This flexibility enables investigations into various O-RAN deployment scenarios, including Open Centralized Unit (O-CU), Open Distributed Unit (O-DU), and Open Radio Unit (O-RU), positioned respectively at regional servers, edge servers, and cell sites.
- **Support for Orchestration and Management (O&M) Functionalities.** The platform must be capable of adding or integrating multiple additional functionalities within the Service Management and Orchestration (SMO) layer. This includes extending rApp platforms, AI/ML Framework (AIMLFW), slicing capabilities, and policy frameworks to cater to evolving needs and requirements.

- **Support Technology Proof-of-Concept (PoC).** The platform should offer sufficient flexibility to accommodate technology PoCs. These PoCs can serve as blueprints for future 6G infrastructure development, allowing for experimentation and innovation.
- **Native AI/ML Framework.** The platform must support highly distributed and native AIMLFW that should encompass cross-domain and federated AI modules, with a focus on enhancing Radio Intelligent Controller (RIC), SMO, and O-Cloud functionalities.
- **Openness and Accessibility.** The platform must enable instantiation of Open Source Software (OSS) components and allow open access by researchers to the internals of the network.

The CCI xG Testbed [1] is a research and testing facility for advanced wireless networking and O-RAN, located in the Washington DC metro area and Virginia Tech (VT) main campus in Blacksburg, Virginia, US (Fig. 1). It is the first of its kind to deploy an end-to-end (E2E) O-RAN test network and provide a Citizens Broadband Radio Service (CBRS) experimentation platform both indoors and outdoors. The testbed was designated by the O-RAN Alliance in 2023 as the first OTIC in the DC, Maryland, and Virginia (DMV) area.

This article provides an overview of the capabilities and functionalities of xG Testbed, including its E2E O-RAN-compliant design and detailed information about the indoor and outdoor sites. By describing the unique design features and architecture of the testbed, our objective is to provide guidance and lessons learned for the development of other O-RAN test facilities worldwide. We propose a natural workflow from lab to in-the-field experimentation and testing, contributing to accelerating O-RAN adoption and deployment, and seeking to produce a more skilled workforce.

RELATED WORKS

The National Science Foundation (NSF) Platforms for Advanced Wireless Research (PAWR) program funded four city-scale testbeds construction across the US: POWDER, COSMOS, AERPAW and ARA [2], all of which have been designated OTIC. However, only two of them (namely, POWDER and COSMOS) have enabled O-RAN testing and experimentation, focusing mainly on O-DU and O-RU. POWDER offers experimentation on a CBRS ecosystem. However, only experimentation on the SDR-based CBRS Device (CBSD) is enabled, without considering the Spectrum Access System (SAS).

ORBIT [3] has a radio-grid ceiling deployment with over 200 USRPs, a large number of which are legacy systems and do not support either new version of the OSS cellular stacks or O-RAN implementation, not allowing users to leverage advanced wireless networking R&D.

Colosseum [4] is a wireless emulator with 256 programmable SDRs (all X310s), each connected to a server. However, it does not operate Over-the-Air (OTA) and is limited to emulation-based experimentation. Arena [5] is an antenna ceiling-grid deployment with 24 USRPs (N210, X310) located on a rack and directly connected to a bare-metal machine. Microsoft has an enterprise-scale O-RAN based testbed [6] located in its Redmond (US) and Cambridge (UK) campuses,

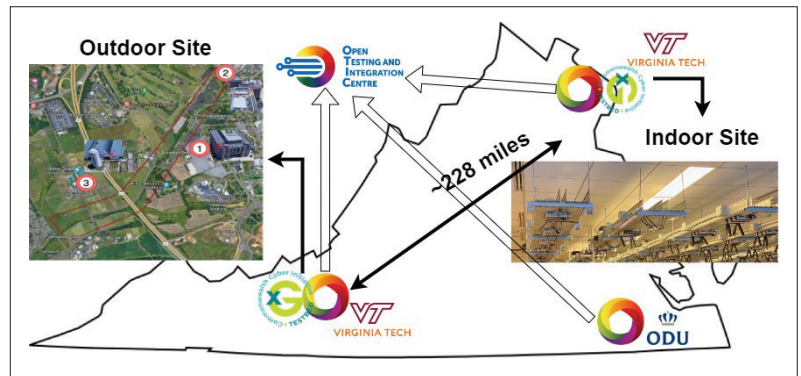


FIGURE 1. CCI xG Testbed Site Locations: Indoor Site at Arlington, VA and Outdoor Site at Blacksburg, VA. The O-RAN logos indicate the three lab locations that are part of the Open Testing and Integration Centre (OTIC).

mainly composed of commercial components and not remotely accessible. SLICES [7] focuses on future and emerging digital infrastructures, with an overall goal of

creating a flexible platform designed to support large-scale, experimental research in networking across Europe. There are some other testbeds in Europe and Asia that are mostly designed to support a specific vertical. For example, 5GinFIRE focuses on VNF experimentation, and \ac{JOSE} has its focus on sensor networks and SDN. In contrast, the xG testbed is designed to offer an open and flexible platform for experimental scenarios spanning multiple application areas. It offers the following unique characteristics compared to the existing testbeds:

- To the best of our knowledge, it is the first platform to enable E2E O-RAN experimentation using SDRs and OSS components, including AIMLFW, Non-RT RIC, Near-RT RIC, RAN (4G and 5G) and UE.~\cite{cci_milcom23}
- It is the only platform worldwide to provide an E2E CBRS ecosystem that offers experimentation capability due to the availability of SDR-based CBSDs and ESC nodes, an OSS SAS (namely OpenSAS), and CBRS PAL as well as FCC experimental licenses.
- It has a radio-grid deployment with over 70 latest-generation USRP (e.g. N310, X310, X410), enabling E2E advanced wireless experimentation with 5G and O-RAN, and capable of operating in different frequency ranges (in FR1, FR2, and FR3).
- It provides a combination of commercial and open-source components that are remotely accessible, and embraces network virtualization by enabling Virtual Machines (VMs) and containers to be dynamically connected to the SDRs.

The remainder of this article is organized as follows. The architecture and two sites of the testbed are presented. A brief description of the OTIC related activities is provided. Furthermore, two sample experiments executed in the testbed are showcased to highlight the capabilities of the testbed, followed by a conclusion.

TESTBED SITES AND ARCHITECTURE

The xG Testbed sites enable a wide range of research in advanced wireless networks and applications, leveraging OpenStack as the foundation for resource O&M. Through this framework, we provide a robust and scalable platform for creating,

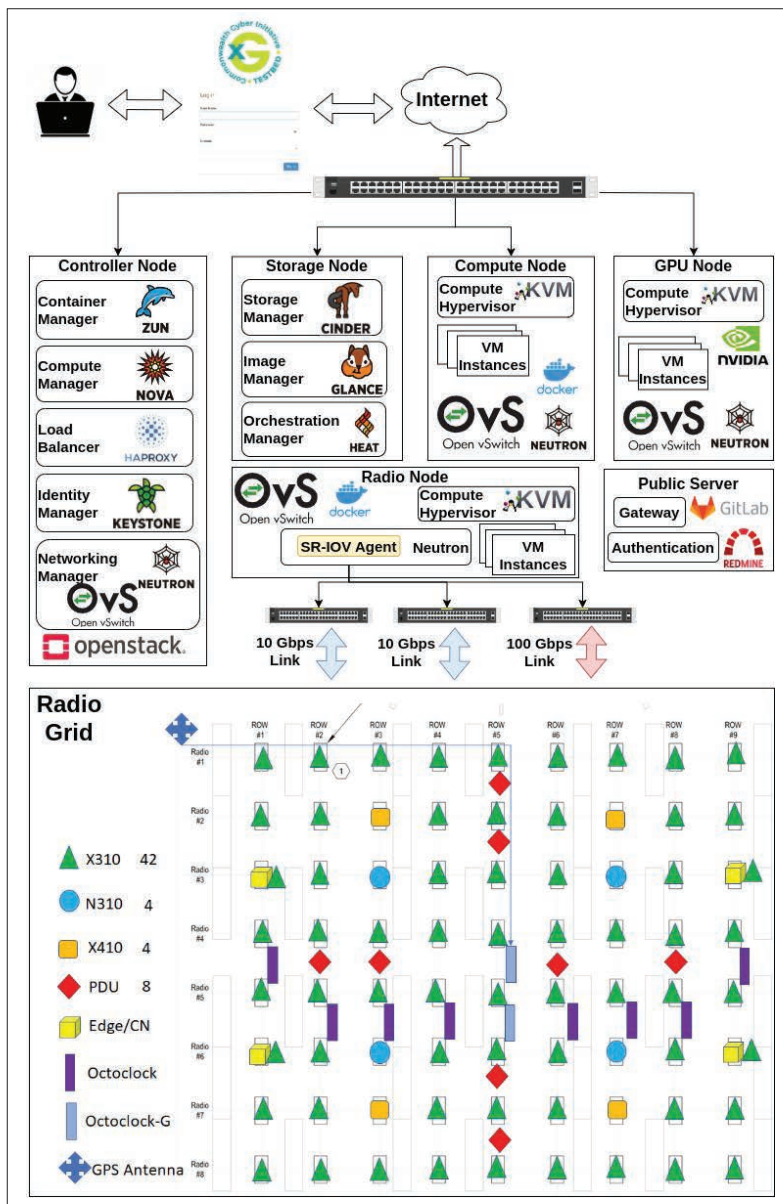


FIGURE 2. Radio-Grid Network topology and system design.

deploying, and managing cloud resources, including general-purpose computing, SDR access, and GPU computing. To enable flexibility for a wide array of projects, we allow both VM and container management, deploying Kubernetes clusters over bare-metal machines. By leveraging an existing open-source platform, we aim to build on top of resilient hardware and flexible scale, while catering specifically to the use-case of provisioning ML and baseband processing capabilities through virtualization.

The testbed infrastructure can be visualized as a pyramid, at the base of which is the core infrastructure, including bare-metal hardware, GPU clusters, SDR grid, antenna, switches, RF and optical fiber cables etc. OpenStack-based O&M provides the second layer of the pyramid, where users can use the resources as they please for experimentation. Users can implement their own 5G/O-RAN building blocks through the various flavors available for RAN, Core Network (CN), Near-RT RIC, and Non-RT RIC that are tested and available on the platform. This level of flexibility allows users

to set up their own experimentation and test cases that they can fine-tune according to their needs. The third layer, forming the apex of the pyramid, is O-RAN testing and experimentation explicitly curated towards OTIC testing, as well as research that can leverage an E2E system based on building blocks available through OpenStack. These are tested and verified to run batches of experiments that enable validation of inter-operability and testing of xApp and rApp software binaries.

Most of the existing testbeds have dedicated compute resources that drive the baseband processing before sending IQ samples to the SDRs. In the xG Testbed, we focus on a virtualization-first approach to enable more flexible experimentation and more efficient resource management while keeping within latency constraints demanded by baseband processing for waveform generation. This allows CPU cores, RAM, and storage on demand to share the rest of the server resources with other experiments, projects, or use cases.

An OpenStack-based portal allows an CCI xG Testbed user to log on remotely through the Horizon dashboard, run experiments and collect data in the testbed's virtual environment. The users can access testbed resources by SSH sessions, that allow them to deploy their own software or use an instantiated software provided by the testbed, and build on them to create their own experiments. The users can also use the Bring-Your-Own-Device (BYOD) service of the testbed by approaching the admin team describing details of the particular device to be integrated to the testbed infrastructure.

INDOOR SITE — RADIO GRID NETWORK

The radio-grid network is located in the VT Research Center (VTRC) in the Washington DC (Fig. 1). It is an E2E open-access SDR-based wireless experimental platform on an indoor ceiling grid, operating on sub-6 and 7.1 GHz. The key characteristics of this site can be summarized as follows (Fig. 2):

- **NEXTG of USRP:** It features a mix-and-match of N310, X310 and X410 USRPs, collectively forming a grid of 72 SDRs. All X310s are equipped with double UB160 for dual-band operation.
- **OTA Operation:** It operates OTA using FCC experimental license which allows operation in any frequency apart from the restricted ones listed in FCC rules Section 15.205(a) [9].
- **Full Synchronization:** The radios are synchronized via eight follower clock distributors connected to two leaders, one of which is connected to a GPS antenna. This enables experimentation and testing on massive MIMO.
- **Virtualization:** Each radio can be dynamically connected to a remotely accessible VM or container.
- **Remote Power Control:** Each radio is connected to a Power Distribution Unit (PDU), allowing energy monitoring and remote access capabilities for users to turn on/off the SDR.
- **Cloudification and Orchestration:** All the radios are connected through 10G multi-mode fiber to an OpenStack-based cloud environment, enabling VM and container orchestration.
- **AI/ML Acceleration Capability:** Three NVIDIA GPUs servers are connected to the OpenStack cloud to support AI acceleration and high performance computing.
- **O-RAN Digital Twin:** It acts as a digital twin of

the E2E O-RAN environment to enable rApp and xApp testing before moving them to a production environment.

OUTDOOR SITE — CBRS STROUBLES CREEK NETWORK

VT was one of the first US universities to acquire CBRS PAL licenses through FCC auction 2020 [10], including four 10-MHz blocks each in Montgomery and Craig Counties, Virginia. Taking advantage of this spectrum, xG Testbed expanded its indoor footprint to include an outdoor site, Stroubles Creek CBRS Network, named after a creek that flows under the deployment area. The production component of our testbed is used for flood monitoring IoT.

The core characteristics of the outdoor site are:

- **Deployment Area:** It covers a 1.5-mile corridor with Base Stations (BSs) deployed in three rooftop locations, shown in Fig. 1.
- **Spectrum:** CBRS PAL and FCC experimental license are used.
- **Two Operational Networks:**
 - A commercial network deployment with “vendor-specific components” and Federated Wireless as SAS service provider
 - An experimental network composed of OSS and SDR-based components including: three CBSD prototypes, Open5GS core, and an OSS SAS developed in-house (named OpenSAS [11]).
- **Open-source SAS (OpenSAS):** It is composed of ESC, incumbent detection and SAS-to-SAS communication.
- **CBSD Prototypes:** 5G srsRAN and SDR-based prototypes operate in Time Division Duplexing (TDD) mode. These include USRP X310 based RF front-end connected to a CommScope VVSSP-360S-F multi-band antenna, an Intel NUC (serving as edge node), a PiKVM and a CyberPower Ethernet PDU with an energy meter for remote operations, and a Dell Switch connecting to the cloud environment located in Arlington, VA. The compute node hosts a CBSD client system, enabling communication with OpenSAS, which manages and orchestrates the CBSD operations (including spectrum access). The prototype built in-lab is fully compliant with 3GPP specifications and is able to register to any SAS provider. Figure 3 shows one of the rooftop deployments with the CBSD prototype. We have tested the 5G srsRAN CBSD by connecting multiple COTS UEs and 5G modems, including OnePlus 8T, OnePlus Nord 5G, Google pixel 6, Telit FN980 5G, Quectel RM510Q-GL M2 5G module with Sycosom programmable SIM cards.
- **E2E CBRS Ecosystem:** to the best of our knowledge this is the first platform to provide an E2E CBRS ecosystem for experimentation.

O-RAN OTIC LAB

In 2023, the CCI xG Testbed was designated by the O-RAN Alliance as an OTIC in the Washington DC metro area, with a vision to serve as a center of excellence to support O-RAN R&D in the DMV area for US government use cases, advance workforce development and accelerate E2E deployment and testing. The center, co-hosted by AT&T, Verizon and Dish Networks, paves the path for a strong collaboration between industry and academia towards O-RAN adoption.

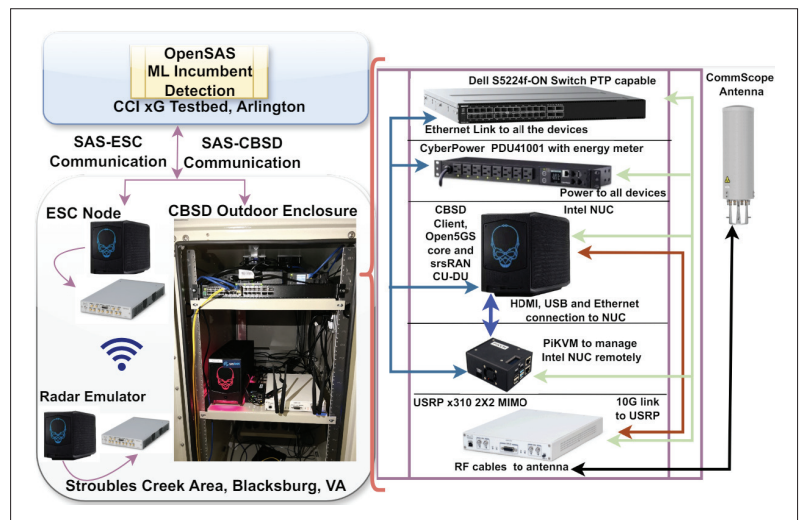


FIGURE 3. CBRS architecture with an outdoor CBSD rooftop deployment at Location 1 (Fig. 1).

The OTIC focuses on performance, conformance, interoperability, and security testing. It is composed of three laboratories located in Virginia as shown in Fig. 1:

- xG Testbed (Arlington, VA);
- Old Dominion University (ODU) (Coastal Virginia);
- VT (Southwest Virginia), which complement each other to realize the OTIC services. The xG Testbed orchestrates and coordinates the OTIC activities (see [12] for details) across the other two labs.

The OTIC is enriched with E2E testing workflow composed by core testing components from Keysight and Rohde & Schwarz. This results in an integrated solution for conformance, performance, and interoperability testing according to O-RAN Alliance specified processes and procedures. The center also serves as a satellite lab for the Acceleration of Compatibility and Commercialization for Open RAN Deployments (ACCoRD) project [13], providing a testbed to support T&E activities funded by National Telecommunications and Information Administration (NTIA). Goals include: Open FrontHaul (OFH) testing between multi-vendor O-DUs and O-RUs; providing an E2E OSS reference O-RAN architecture for OFH testing; serving as the nucleus for the Cloudified Federated Lab as a Service model; and providing coordination and technical support for neutral host facilities across US and abroad.

SAMPLE EXPERIMENTS

Two sample experiments demonstrating some of the critical capabilities and functionalities of the two main xG Testbed sites, that can be leveraged towards more advanced experimentation or customized scenarios follows.

EXPERIMENT 1: HETEROGENEOUS COEXISTENCE BETWEEN WI-FI AND 5G NR-U IN THE UNLICENSED SPECTRUM USING SDRs

Background: In light of licensed spectrum scarcity, spectrum sharing is one of the key enablers for the next generation of wireless communications. The unlicensed spectrum provides an avenue to increase spectral resources at the cost of additional latency incurred due to the Listen-Before-Talk (LBT) mechanism required for channel operation with other devices in the shared

Dynamic Frequency Selection (DFS) channels shared with radar and weather devices are vacant and through the experimental FCC license, the xG Testbed can assume control for experimentation.

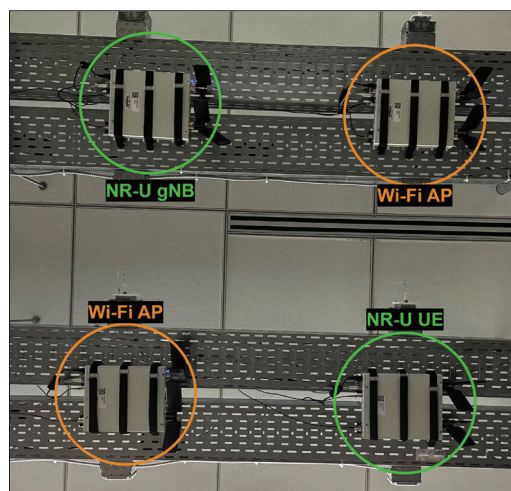


FIGURE 4. Indoor setup to test 5G NR-U and 802.11a Wi-Fi coexistence using only software-defined radios in the 5 GHz U-NII band.

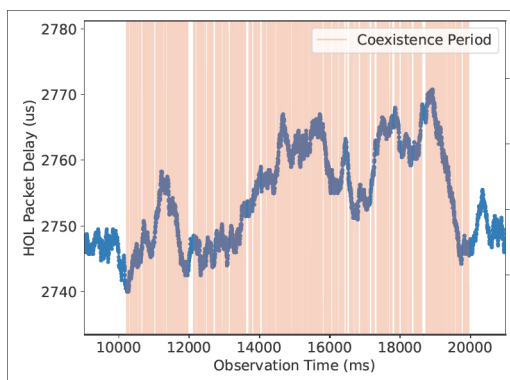


FIGURE 5. Moving Average HOL Packet Delay for NR-U downlink transmission with and without Wi-Fi traffic.

spectrum. 3GPP introduced the New Radio Unlicensed (NR-U) specifications in Release 16, which extend cellular communications into the unlicensed spectrum, enabling Mobile Network Operators (MNOs) an opportunity for a cost-effective increase in network capacity [14].

Objective: With the help of the indoor radio grid at CCI and the flexibility of prototyping multiple SDRs with different Radio Access Technologies (RATs) through remote access, we explored the scenario of channel access by introducing OFDM Wi-Fi transmissions between two devices operating as an NR-U RAN and UE, respectively, by leveraging the LBT mechanism over the OAI platform. Attempting LBT over a full 5G stack has never been explored before, and this work is one of the first that attempts to define the current state of how software-based LBT can behave alongside an E2E 5G RAN and CN setup and provide a PoC of this functionality running with OAI.

System Model: To enable unlicensed spectrum access, we incorporate the Category-3 LBT [14], defined for Physical Downlink Shared Channel (PDSCH) transmissions, running on top of the OAI 5G RAN stack connected to a 5G CN and OpenAirInterface (OAI) UE. In this work, we focus only on downlink traffic and, hence, do not equip the UE with LBT and instead rely on the BS to allow Primary Uplink Control Channel (PUCCH) transmissions within the Channel Occupancy Time (COT). To introduce contention, we

transmit OFDM signals with two Wi-Fi transmitters operating in the same channel.

Methodology: We setup an NR-U BS, to which an NR-U UE is attached, to allow random access without interference followed by OFDM transmissions based on the IEEE 802.11a PHY layer specifications for 10 seconds after the PDSCH channel is setup. We operate this co-existence scenario in the IEEE Channel 100 of the 5 GHz unlicensed band (n46), which has a center frequency of 5.5 GHz. Since we operate OTA without channel emulation, the U-NII-1 and U-NII-3 channels are occupied with incumbent Wi-Fi transmissions. However, the U-NII-2c and U-NII-2a channels (which includes the 5.5 GHz channel) which are Dynamic Frequency Selection (DFS) channels shared with radar and weather devices are vacant and through the experimental FCC license, the xG Testbed can assume control for experimentation. In order to minimize the backoff duration to avoid skipping too many slots, we set the Contention Window (CW) to 3 with a sensing slot duration of 500 μ s to align with the samples-per-packet rate configured by OAI with the UHD driver.

Results: In Fig. 5, we characterize the average Head-of-Line (HOL) packet delay of incoming Radio Link Control (RLC) packets in NR-U and observe that it increases by around 20 μ s during the coexistence period with Wi-Fi transmitters for over 10 seconds before dropping down after the Wi-Fi transmissions cease. The period highlighted in red is determined by the received energy breaching the energy detection threshold. Since we are operating only with a single NR-U gNB and UE along with two Wi-Fi transmitters, the packet delay is affected primarily due to the additional offset of having to wait for idle state rather than the channel capacity limits due to the small number of devices under test. Further experimentation can be conducted by incorporating FPGA-based LBT and interfacing it with the UHD driver running on OAI which will enable more accurate specification-compliant LBT and leveraging larger array of devices for a deeper study into the co-existence behavior using real-world hardware.

EXPERIMENT 2: ML-BASED INCUMBENT DETECTION IN CBRS

Objective: To demonstrate the xG Testbed's spectrum sharing research capabilities through ML-based incumbent detection in E2E experimental CBRS networks.

System Model: Figure 3 shows the experimental setup with four main components: OpenSAS for spectrum management (along with ML-based incumbent detection), and SDR-based ESC node, incumbent emulator and CBSDs.

For the indoor scenario, the radar emulator and the ESC node are deployed in the radio-grid, and for the outdoor scenario, both are deployed in Blacksburg, VA. The CBSDs are deployed in both indoor and outdoor sites. Figure 3 also includes the outdoor enclosure deployed at outdoor site Location 1 (Fig. 1) with CBSD prototype.

Methodology: We extend our previous work on ML-based incumbent detection in an experimental CBRS network [11], where we presented the indoor OTA incumbent detection accuracies and E2E detection latencies using incumbent (radar) dataset from (National Institute of Standards and Technology (NIST)). To calculate detection accu-

racy of the ML model, the incumbent signal is transmitted using an incumbent (radar) emulator in an outdoor OTA environment, the incumbent emulator iterates through the radar signal dataset [11] and transmits radar signal in a loop.

The ESC node monitors the CBRS band and upon detection of activity in any channel in the CBRS band, it sends the captured IQ samples to the ML model running inside OpenSAS which classifies it as either incumbent or unknown (non-incumbent) signal. The OpenSAS, upon detecting the incumbent in a channel, terminates the grants (permission to use the channel) of lower-tier users, i.e., PAL and General Authorized Access (GAA) CBSDs. The CBSDs stop their transmission when the grant is terminated. The channel occupancy (in the CBRS band), CBSD status information, and sensing views are available on OpenSAS dashboard [11]. If a non-incumbent signal is detected, the sensing view displays as *unknown* for the corresponding channel, and the grants for lower-tier users operating in that channel are not terminated.

Results: We evaluate the performance of OTA ML-based incumbent detection in the outdoor environment and compare the OTA detection accuracy with indoor results that we demonstrated in [11]. Additionally, we measured the E2E latencies in incumbent detection, which are critical for other CBRS like Dynamic Spectrum Sharing (DSS) use-cases as recommended in [15]. Table 1 compares indoor and outdoor incumbent detection accuracies and latencies (averaged over 500 samples). The real-world effect due to indoor and outdoor channels, and hardware impairments adversely affects the OTA detection accuracy. As compared to the training and validation accuracies, we saw a decrease in the OTA detection accuracies for indoor scenario due to these real-world effects [11]. The outdoor detection accuracy for all three Signal-to-Noise Ratio (SNR) levels are lower than those observed indoors. This is because the ML model was trained with the indoor OTA-captured radar signal, and channel conditions changed.

The overall detection latency is the sum of individual latencies in Table 1 and includes time required to:

- Move the center frequency to the detected channel,
- Change the sampling rate (srate) to capture the detected channel (capturing IQ samples),
- Compute FFT,
- Receive samples
- Send them to OpenSAS for predicting the outcome (HTTPS request time)
- Return to the center frequency for observing the entire spectrum again.

Except for HTTPS request time, all other latencies are similar, as the computation is same for both indoor and outdoor experiments. Only the distance between the OpenSAS server (Indoor site) and the ESC node (Indoor and Outdoor sites) changes. Hence the HTTPS latency between OpenSAS server and ESC node increases when the latter is located in Blacksburg (Outdoor site).

OTHER SAMPLE EXPERIMENTS

The testbed supports various experimental research in cellular and O-RAN technologies including but not limited to:

		Indoor [11]		Outdoor	
		Incumbent SNR			
Detection Accuracy (%)	20–30 dB	68.92		59.45	
	30–40 dB	79.56		77.65	
	40–50 dB	85.35		81.75	
Latency (ms)	Types	Mean	SD	Mean	SD
	Frequency shift	116.84	0.9947	116.798	1.05
	Frequency return	115.545	2.933	115.955	3.05
	srate change	0.1832	0.042	0.184	0.041
	Sample receive	0.01	0.013	0.01	0.039
	HTTP request	227.375	129.842	944.905	510.395
	FFT Computation	0.0031	0.2	0.003	0.196

TABLE1. Indoor and outdoor incumbent detection accuracies and latencies (Standard Deviation (SD)).

1. *NEXTG Experimental Cellular Networks:*
 Opensource 4G and 5G deployment using srsRAN and OAI to enable experiments like channel characterization, joint wireless, wired and edge-cloud continuum, massive MIMO, mmWave (28GHz) beamforming, URLLC, E2E network slicing, traffic steering with heterogeneous RATs. When enabling such experiments, the users may require to modify the OSS cellular stacks (i.e., srsRAN, OAI) and thereby contribute to the OSS community. The testbed hosts the srsRAN North-America lab and continuously works with Radio Systems (SRS), for example, to enable the modification of key parameters in the srsRAN via xApps. It also partners with National Instruments (NI) to create a SDR-based (X410) O-RU.
2. *Spectrum Sharing:*
 - Homogeneous and heterogeneous DSS,
 - Priority protection and interference management,
 - Secondary coexistence in CBRS and other multi-tier spectrum sharing ecosystem,
 - Carrier aggregation (both licensed, shared, and unlicensed spectrum),
 - 5G NR-U contention management & coexistence with Wi-Fi.
3. *O-RAN Research and Testing:*
 - E2E O-RAN ecosystem for developing and testing dynamic RAN optimization such as resource allocation, energy efficiency, QoS and QoE optimization etc.,
 - AI/ML based xApp and rApp development,
 - xApp conflict mitigation (direct and indirect)
 - Distributed control, orchestration and optimization of disaggregated RAN components,
 - E2E AI/ML data pipeline with inference at RICs , and training at AIMLFW from O-RAN Software Community (SC),
 - Security evaluation of O-RAN interfaces,
 - OFH Conformance and Interoperability Testing

CONCLUSION

In this article, we presented the first E2E O-RAN-compliant experimental testbed, the CCI xG Testbed, that is SDR-based and comprises of OSS and commercial components. With both indoor and outdoor components as its integral elements, the platform provides an O-RAN T&E infrastructure as OTIC, paving the path to accel-

erating O-RAN. The testbed architecture provides a fully programmable and remotely accessible living lab for experimentation and testing with 5G-and-beyond solutions. Future works include providing a digital-twin ecosystem for rApp and xApp assessment, leveraging OTIC E2E security testing, evolving the AI-native aspects of the testbed (for example, by adding transfer & federated learning functionalities), working with other testbeds towards cross-testbed and OTIC federation, and enabling E2E spectrum sharing and 5G NR-U experimentation. Additionally, there are also plans to establish the FCC innovation zone, enable VT drone park for UAV experimentation and leverage partnership with Linux Foundation to fully deploy Open Network Automation Platform (ONAP) and implement LF Edge framework.

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BIOGRAPHIES

ALOIZIO DA SILVA (aloiziops@vt.edu) is the CCI xG Testbed Director and Electrical and Computer Engineering (ECE) faculty at VT. He earned his B.S., M.S., and Ph.D. from Pontificia Universidade, Universidade Federal, and Instituto Tecnológico de Aeronáutica, respectively. He was a Technical Project Manager for NSF PAWR and worked in US-IGNITE, University of Bristol, and JPL-NASA. His research interests include O-RAN, Spectrum Sharing, AI for Wireless and VNF.

MAYUKH ROY CHOWDHURY (mayukhrc@vt.edu) is a Postdoctoral Researcher at CCI, VT. He received his Ph.D in Electrical Engineering at the Indian Institute of Technology (IIT) Delhi in 2022 and M.Tech at IIT Patna in 2016. His research interests include AI for Wireless, O-RAN, Spectrum Sharing etc. He has worked in 6G Lab at Samsung Research and TCS Innovation Labs.

ADITYA SATHISH (saditya@vt.edu) is a Ph.D. student in ECE at VT. He focuses on improving 6G service quality, virtualization and SDN, especially within the O-RAN framework. He worked at Qualcomm, and his expertise includes IEEE 802.11, Embedded Software, and 5G/6G networks.

ASHEESH TRIPATHI (asheesh@vt.edu) is a Ph.D. student in ECE at VT, with a focus on spectrum sharing for next-gen wireless networks. Asheesh has industry experience in testbed design and deployment, cellular/ O-RAN T&E, during his tenure at Verana Networks, Parallel Wireless, and Idea Cellular. He received his MSEE degree from the University of North Carolina at Charlotte.

SCOTT MIDKIFF (midkiff@vt.edu) is a Professor of ECE at VT. Midkiff previously served as the university's CIO, ECE department head, and a program director at the NSF. His research interests include wireless networks, IoT, and cybersecurity. He received his B.S.E. and Ph.D. degrees from Duke University and his M.S.E.E. degree from Stanford University.

LUIZ DASILVA (aldasilva@vt.edu) is the Executive Director of the CCI and the Bradley Professor of Cybersecurity at VT. Previously, he was the Professor of Telecommunications at Trinity College Dublin and Director of CONNECT. His research focuses on adaptive resource management in wireless networks, cognitive radio networks, and game theory applications in wireless settings.